Instruct Models and Post-Training

Instruct Language Models

- The original definition of a language model was a probabilistic model that defines a distribution over finite strings, given a vocabulary
  - Equivalently, the prediction of the next word (token) given a prefix

- Many tasks can be cast as (iterative) next word prediction
  - Therefore, effectively performing next word prediction may allow zero-shot or few-shot learning for many tasks
  - OpenAI, with their GPT3 model (Brown et al., 2020), has shown that optimizing for next word prediction yields a strong model for few-shot and zero-shot learning

Instruct Language Models

- Instruct language models have redefined the notion of a language model to a system that obeys instructions
  - Initially, instructions to produce text
  - More recently, instructions to perform actions (agents)

- Instruct models are based on pretrained, next word prediction models that serve as <u>initialization</u> to a subsequent stage, sometimes called post-training

Instruction Following vs. Next Word Prediction

Main differences:

1. Instruction following is an interaction → requires adherence to the implied intent of the user
2. Instruction following usually imposes safety constraints / guardrails
3. Next word prediction is a density estimation problem → we seek to give probability to all ”good” completions

Instruction Following vs. Next Word Prediction

Most medieval European scholars believed that the Earth was …

meta-llama/Llama-3.1-8B

- flat, and they believed it was flat because they
- a flat disk, surrounded by water
- flat
- at the center of the universe, but this belief
- at the centre of the universe and that the stars

Olmo 3 32B Instruct

Most medieval European scholars believed that the Earth was **at the center of the universe**, with the Sun, Moon, planets, and stars revolving around it. This view is known as the **geocentric model** and was based largely on the ideas of the ancient Greek philosopher Ptolemy, whose work *Almagest* was highly influential throughout the Middle Ages.

Post-Training

- The details of post-training are often undisclosed by the companies that develop these models

- However, it is known that the post-training process has an initial step where the ”core skills” are defined, and that the acquisition of these skills is achieved using these two components:
    1. Supervised fine-tuning (SFT)
    2. Reinforcement learning using policy gradient methods*

* There are also newer methods that use other algorithms

TÜLU 3: Pushing Frontiers in Open Language Model Post-Training. Lambert et al., 2024

Core Skills in TÜLU 3

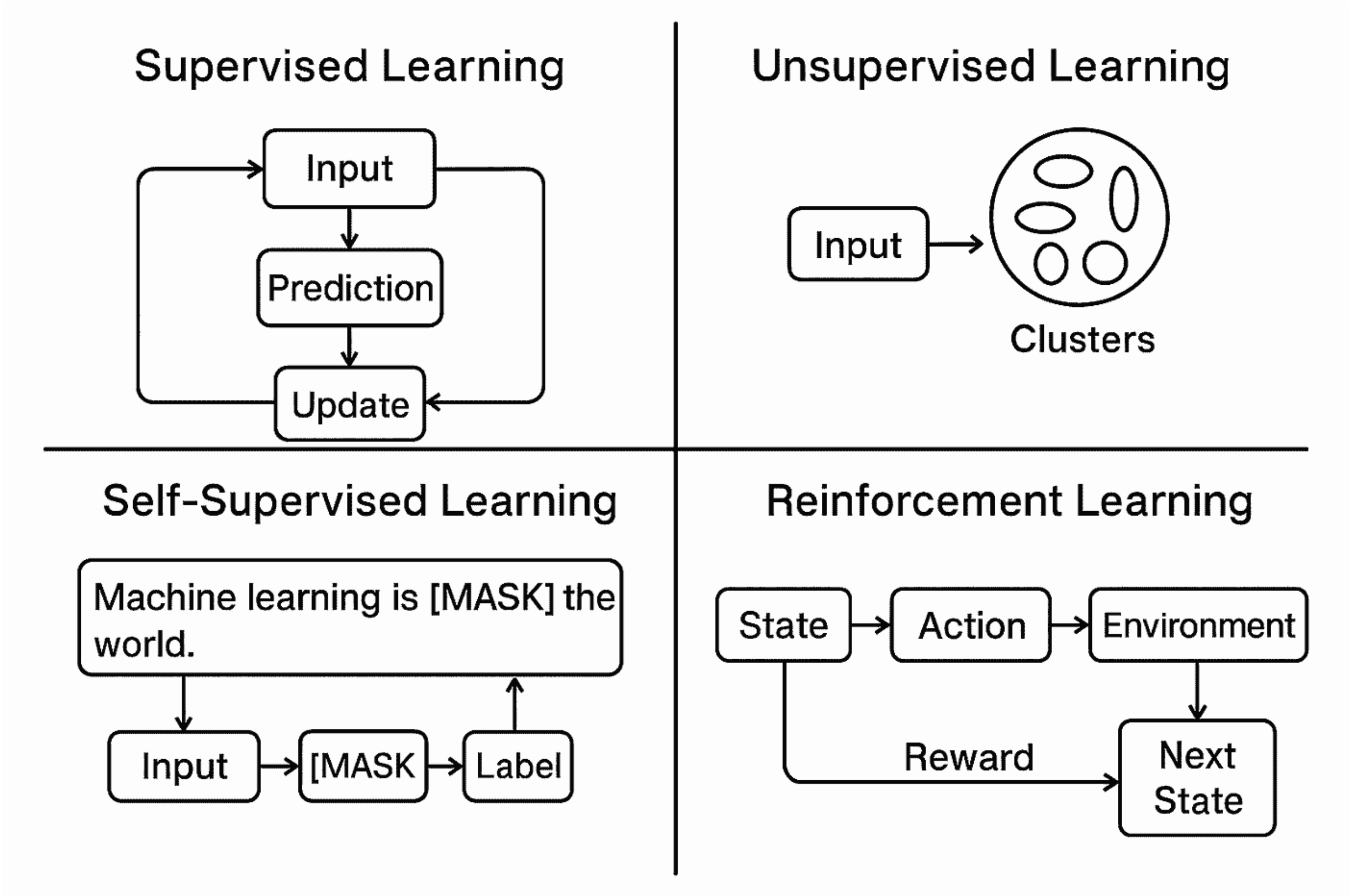
- Other models may share or diverge from this set
- Curation of prompts for these ”core skills” is a combination of data curation and synthesis

Supervised Fine-Tuning (SFT)

- The simplest training method employed
- Identical to fine-tuning a model for any other task
- Training proceeds by feeding the decoder with the prompt and training it to predict the first word,
  - then the prompt and the first word, training it to predict the second
  - then the prompt and the first two words, training it to predict the third
  - …

- Main shortcomings for this method: exposure bias and inability to learn from negative examples

Bird's Eye View of Reinforcement Learning



Reinforcement Learning from Human Feedback (RLHF)

- Seeking to address the exposure bias and to leverage negative examples, we define a language model $\pi(\cdot|x)$, where x is some prefix, by initializing it from the SFT model
  - π is called the "policy"
- We assume that there is some real function *r(x,y)* where *y* is a continuation of *x*, that scores the different continuations, such that the better *y* is, the higher *r* is
- We seek to optimize the expected reward:

$$\mathbb{E}_{\pi}\left[r\right] = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x)}\left[r(x,y)\right]$$

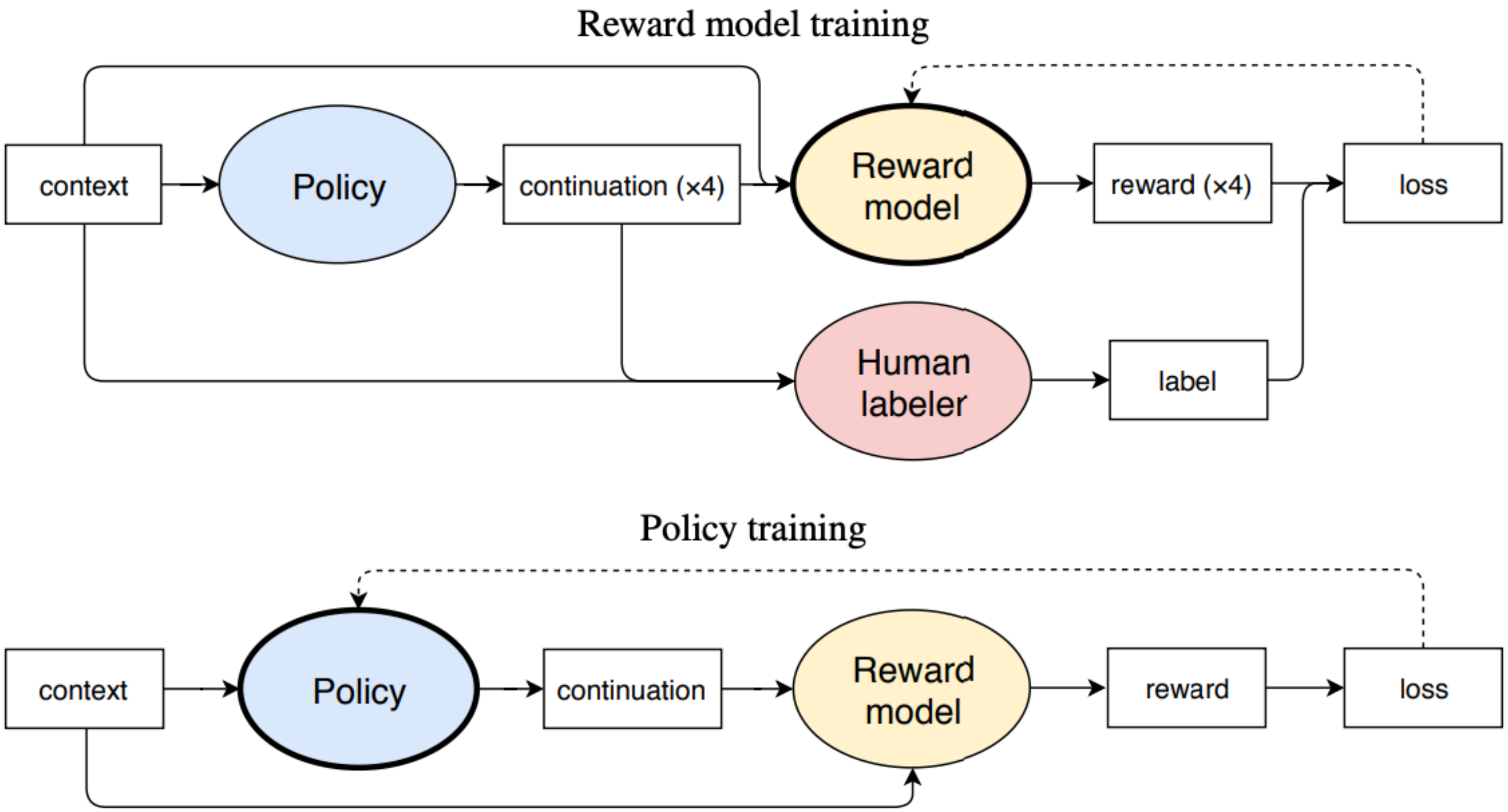
Ziegler etl al., 2020. Fine-Tuning Language Models from Human Preferences

Reinforcement Learning from Human Feedback (RLHF)

- The difficulty is that *r* is unknown to us, so we need to train it as well

- We assume that we have a sample of prefixes *x* (instructions) and for each, we have 4 options, sampled from our current LM p. We further assume annotation of which of the 4 options is best.
- We train p to increase the expected reward by changing its probability. In parallel, we train r to conform to the preferences of the humans.
- A penalty term keeps π from diverging too much from the SFT LM

Reinforcement Learning from Human Feedback (RLHF)



Results

- As mentioned last class: very difficult to evaluate
  - Many benchmarks used, testing for a wide variety of capabilities

Training for MATH does not transfer well to Deepmind Math. One difference between MATH, which is part of our development evaluations and DeepMind Math, which is in our unseen evaluations is that the former often

Models generally overfit to IFEval. We find that there is a significant difference between performance on IFEval and IFEval-OOD of all the models, even though we created the latter to be structured very similar to the original dataset, just with a disjoint set of constraints. We observe that instruction following with verifiable constraints is a

Instruction following performance varies across categories. We observe that the relative performance of TÜLU 3 models on AlpacaEval is different from that on HREF. This may be explained by the fact that instruction following is a highly diverse task, and the distributions of HREF and AlpacaEval may differ, with some categories of instructions

TÜLU 3: Pushing Frontiers in Open Language Model Post-Training. Lambert et al., 2024

Agents

- The most recent trend in language models are models that not only follow instructions by generating text, but also by performing actions for the user
- Actions include:
  - Interacting with web APIs and other tools
  - Navigating through websites to fill out forms or extract information
  - Writing code and running it
  - …
- More about it in advanced courses (e.g., Advanced NLP, a course by Tom Hope)

Conclusion

- Instruct models have brought about a new definition of language models
  - Not continuing prefixes, but following instructions
  - More recently, also performing actions
- A fast-moving and challenging field
  - Progress, however, is very difficult to measure and at times, in the eyes of the beholder